ECA0806-Analog & Digital Communication

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Experiment-10

PERFORMANCE ANALYSIS OF FM SYSTEM UNDER GAUSSIAN WHITE NOISE USING MATLAB

Program (Matlab Code):

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;          % Message frequency
fc = 5000;         % Carrier frequency
kf = 100;          % Frequency sensitivity

% Message and Carrier Signals
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);

% FM Modulation
fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);

% Add Gaussian White Noise manually (10 dB SNR) without Communications Toolbox
snr_dB = 10;
signal_power = mean(fm_sig.^2);
noise_power = signal_power / (10^(snr_dB / 10));
noise = sqrt(noise_power) * randn(size(fm_sig));
noisy = fm_sig + noise;

% Display
figure;
subplot(4,1,1)
plot(t, m)
title('Message Signal')
grid on

subplot(4,1,2)
plot(t, c)         % Fixed: now correctly plots the carrier signal
title('Carrier Signal')
grid on
```

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```
subplot(4,1,3)
plot(t, fm_sig)
title('FM Signal')
grid on

subplot(4,1,4)
plot(t, noisy)
title('FM Signal with Gaussian White Noise')
grid on
```

Output Waveform :

